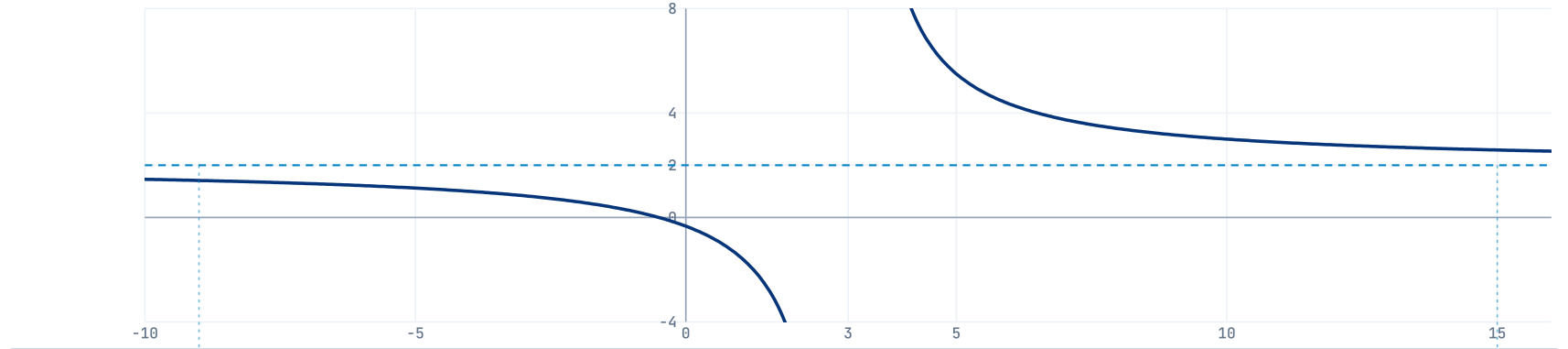
A HORIZONTAL ASYMPTOTE, BOTH ENDS

Profile of the example $f(x) = \frac{2x+1}{x-3}$

2 LIMIT AT ±∞ · 0 CROSSINGS



KEY POINTS

1 ASYMPTOTE
y = 2, both ends

GLOBAL

LIMITS
$\lim_{x\to\pm\infty} f(x) = 2$

GAP TO THE LINE
$f(x) - 2 = \frac{7}{x-3}$

SIDE
above for $x > 3$, below for $x < 3$

CROSSINGS
none
$\frac{7}{x-3}$ is never zero